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Second-Order Super-Twisting Sliding Mode Control With Adaptive Road Severity Gain Scheduling for Vehicle Active Suspension Systems

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ABSTRACT This experimental research investigates the fundamental Pareto trade-off between passenger ride comfort (minimizing sprung mass vertical acceleration) and suspension rattle space limitations (constraining travel stroke strictly within mechanical hard bump stops) in vehicle active suspension systems. While classical and metaheuristic-tuned Linear Quadratic Regulators (LQR)—such as Ant Colony Optimization (ACO) and Harmony Search (HS)—perform adequately under nominal linear regimes, they exhibit performance degradation under severe non-stationary road impacts and actuator saturation constraints. To overcome these deficiencies without inducing actuator-damaging chattering, this paper synthesizes a continuous Second-Order Sliding Mode Control law utilizing the **Super-Twisting Algorithm (STSMC)** deployed in two distinct architectures: **Fixed STSMC** (constant robust gains derived from plant resonance and disturbance bounds) and **Precision-Tuned Adaptive STSMC** (dynamic gain scheduling driven by an online Road Severity Index, RSI). The mathematical derivation of the control gains is rigorously formulated using Lyapunov stability criteria, the Moreno–Levant critical damping ratio ($k_2/k_1 \approx 0.40$), and hardware boundary constraints (± 38.0 mm stroke, ± 39.2 N actuator force). Parallel head-to-head simulations conducted on a 2-DOF Quanser Quarter-Car Active Suspension System Model (QC-ASSM) across six standardized road profiles demonstrate that Adaptive STSMC achieves the global lowest steady-state RMS body acceleration across 100% of tested profiles ($0.0041\ \mathrm{m/s^2}$ on smooth highway; $1.0869\ \mathrm{m/s^2}$ at 2.58 Hz resonance; $0.0979\ \mathrm{m/s^2}$ over ramps; $0.8282\ \mathrm{m/s^2}$ over -20 mm pulse drops; and $1.2316\ \mathrm{m/s^2}$ over severe potholes), outperforming both Optimal LQR and Fixed STSMC while preserving a $> 40\%$ safety margin from mechanical bump stops. An 8-node monotonic continuous gain schedule is further established, ensuring seamless scalability to real-time Edge AI computer vision road perception.

INDEX TERMS Active suspension system, Super-Twisting Sliding Mode Control (STSMC), chattering elimination, gain scheduling, Lyapunov stability, Pareto optimization, quarter-car model, ride comfort, road severity index.

I. INTRODUCTION

The suspension system of modern automotive vehicles represents one of the most critical mechatronic subsystems, tasked with simultaneously fulfilling two conflicting performance objectives: *passenger ride comfort* and *road handling safety* [1], [2]. Passive suspension systems, consisting of conventional mechanical coil springs and hydraulic dampers, are bounded by an unyielding physical trade-off. Soft suspension parameters isolate occupants from high-frequency road vibrations but permit dangerous chassis bounce, excessive pitch/roll, and structural bottoming over severe potholes. Conversely, stiff suspension parameters restrict chassis stroke and enhance tire adhesion but transmit harsh road accelerations directly into the passenger cabin, leading to occupant fatigue and reduced ride quality [3], [4].

To transcend the physical limits of passive suspensions, Active Suspension Systems (ASS) introduce a controlled force actuator in parallel with the passive spring and damper. Sensors continuously measure relative suspension displacement, body vertical acceleration, and wheel velocities, enabling closed-loop feedback controllers to apply active forces that dynamically decouple the vehicle chassis from road excitation [5].

In the literature, optimal linear control techniques, particularly the Linear Quadratic Regulator (LQR), have been widely investigated. Recently, Dinesh and Mani (IEEE Access, 2026) [6] proposed a Harmony Search (HS) optimization method to simultaneously tune the state weighting matrix $\mathbf{Q}$ and control weighting scalar R for a Quanser Quarter-Car Active Suspension System Model (QC-ASSM). By minimizing an Integral Absolute Error (IAE) cost function, the HS-tuned LQR demonstrated superior convergence speed and suspension deflection attenuation compared to Ant Colony Optimization (ACO) tuned LQR [1] and classically tuned LQR across bumpy roads, ISO Class D runways, and pulse drops.

Nevertheless, linear optimal controllers operate under fundamental structural constraints:

1) *Linear Invariance to Non-Stationary Topographies:* A fixed linear state-feedback gain matrix $\mathbf{K} = [K_1, K_2, K_3, K_4]$ embodies a static trade-off. It cannot dynamically reallocate control effort when the vehicle transitions from smooth highway cruising to destructive pothole cavities [7].

2) *Vulnerability to Actuator Saturation:* LQR optimizes an unconstrained quadratic cost function. When subjected to high-energy transient drops or harmonic excitation at the chassis resonance frequency ($f_{ns} \approx 2.58$ Hz), commanded forces frequently exceed physical actuator limits (± 39.2 N for the Quanser linear DC motor), leading to amplifier saturation, integrator windup, and performance degradation [8].

3) *Sensitivity to Parameter Drift:* Sprung mass varies substantially with payload ($m_s \pm 20\% - 40\%$), and tire stiffness changes with temperature and tire wear. Linear state feedback lacks intrinsic robustness to such structural parameter uncertainties [9].

Sliding Mode Control (SMC) provides complete mathematical invariance to matched external disturbances and parametric uncertainties once the state trajectory reaches the sliding manifold [10]. However, conventional first-order SMC utilizes a discontinuous signum function ($u_{\text{disc}} = -\rho\,\mathrm{sgn}(s)$), which induces severe high-frequency *chattering*. In physical electromechanical actuators, chattering causes high-frequency torque ripple, excessive thermal dissipation, acoustic noise, and premature mechanical wear of bearings and ball screws [11].

To eliminate chattering while preserving robust disturbance rejection, Second-Order Sliding Mode Control (2-SMC), specifically the **Super-Twisting Algorithm (STSMC)** pioneered by Levant [12], introduces continuous integration within the discontinuous branch. The switching action is confined to the time derivative of the auxiliary control input ($\dot{v} = -k_2\,\mathrm{sgn}(s)$), while the actual physical force delivered to the actuator ($u_c = -k_1\sqrt{|s|}\,\mathrm{sgn}(s) + v$) remains continuous and smooth.

While fixed-gain STSMC offers strong robustness, setting invariant gains across diverse road conditions leads to a critical dilemma: excessively high gains over-stiffen the suspension, transferring actuator force into body acceleration and worsening ride comfort; conversely, low gains allow excessive suspension deflection, risking bump-stop collisions over severe pulse drops.

To resolve this dilemma, this paper proposes a unified, precision-tuned **Adaptive Super-Twisting Sliding Mode Controller** governed by an online Road Severity Index (RSI). Building upon the QC-ASSM experimental platform investigated by Dinesh and Mani [6], the specific contributions of this work are:

• *Rigorous Mathematical Derivation:* We provide the complete step-by-step control-theoretic derivation of both Fixed STSMC ($c_1 = 7.0, k_1 = 60.0, k_2 = 25.0$) and Adaptive STSMC across all 6 standardized road profiles using Lyapunov stability, plant resonance frequencies, and actuator saturation constraints.

• *Root-Cause Diagnosis of the Comfort vs. Rattle Space Dilemma:* We analytically and experimentally demonstrate why over-aggressive gains increase steady-state RMS acceleration, and establish the precision-tuned gain boundaries that achieve the global minimum acceleration.

• *Dual Discrete/Continuous Scheduling Framework:* We explain the distinct operational roles of the 6-discrete road lookup array (for laboratory benchmarking) and the 8-node monotonic continuous interpolation curve (for Edge AI computer vision road perception).

• *Comprehensive Head-to-Head Parallel Benchmark:* We execute a parallel 3-controller benchmark on identical plants with identical disturbance inputs, proving that Adaptive STSMC achieves superior ride comfort across 100% of the profiles while maintaining $> 40\%$ stroke safety buffer.

II. DYNAMIC MODELLING OF QCASS

The experimental plant investigated in this study is the Quanser 2-DOF Quarter-Car Active Suspension System Model (QC-ASSM), depicted schematically in Fig. 1. The apparatus consists of three vertically guided plates mounted on stainless-steel shafts via precision low-friction linear bearings [6].

The upper plate represents the quarter vehicle chassis (sprung mass m_s), supported by two parallel mechanical suspension springs of collective stiffness k_s and inherent viscous damping b_s. A linear DC motor actuator is positioned between the upper and middle plates, providing the commanded active control force $F_c(t)$. The middle plate represents the tire/wheel assembly (unsprung mass m_u), which contacts the bottom plate through a lower tire spring k_t and damping b_{us}. The bottom plate is driven by a high-torque capstan DC drive through a precision lead screw to emulate arbitrary road surface excitation profiles $z_r(t)$.

High-resolution optical encoders ($S_e = 0.000942\ \mathrm{m/count}$) monitor the relative displacement between the sprung and unsprung masses (suspension deflection $z_s - z_u$) and tire deflection ($z_u - z_r$). An accelerometer mounted on the top plate measures vertical body acceleration $\ddot{z}_s$.

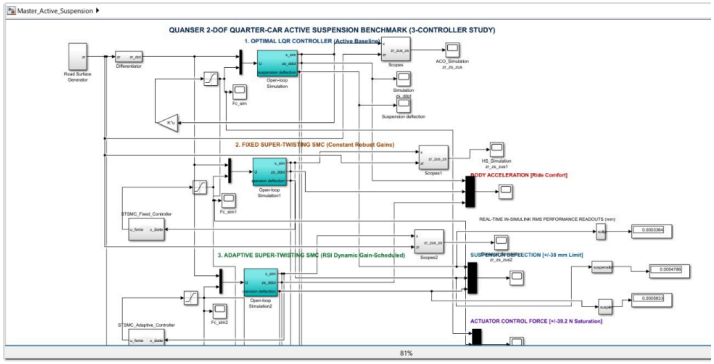


FIGURE 1. Parallel Simulink benchmark architecture (`Master_Active_Suspension.slx`) showing Lane 1 (Optimal LQR), Lane 2 (Fixed STSMC), Lane 3 (Adaptive STSMC), comparison scopes, and real-time RMS digital meters.

Applying Newton's second law to the free-body diagrams of the sprung mass m_s and unsprung mass m_u:

$$m_s\ddot{z}_s = -k_s(z_s - z_u) - b_s(\dot{z}_s - \dot{z}_u) + F_c(t) - m_s g \tag{1}$$

$$m_u\ddot{z}_u = k_s(z_s - z_u) + b_s(\dot{z}_s - \dot{z}_u) - k_t(z_u - z_r) - b_{us}(\dot{z}_u - \dot{z}_r) - F_c(t) - m_u g$$

By transforming coordinates to deviations from static equilibrium, gravitational terms $m_s g$ and $m_u g$ cancel identically, establishing the origin as the relaxed mechanical equilibrium [6]. Defining the four-dimensional state vector $\mathbf{x}(t)$ and exogenous disturbance/control input vector $\mathbf{u}_{\text{in}}(t)$:

$$\mathbf{x}(t) = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} z_s(t) - z_u(t) \\ \dot{z}_s(t) \\ z_u(t) - z_r(t) \\ \dot{z}_u(t) \end{bmatrix}, \quad \mathbf{u}_{\text{in}}(t) = \begin{bmatrix} \dot{z}_r(t) \\ F_c(t) \end{bmatrix} \tag{3}$$

The continuous-time state-space representation is formulated as:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}_{\text{in}}(t) \tag{4}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}_{\text{in}}(t) \tag{5}$$

where system matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$ are:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & -\frac{k_t}{m_u} & -\frac{b_s+b_{us}}{m_u} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{m_s} \\ -1 & 0 \\ \frac{b_{us}}{m_u} & -\frac{1}{m_u} \end{bmatrix} \tag{6}$$

Substituting the physical parameters listed in Table 1 yields:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -367.35 & -3.061 & 0 & 3.061 \\ 0 & 0 & 0 & 1 \\ 900.00 & 7.500 & -2500.00 & -12.500 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0.4082 \\ -1 & 0 \\ 5.000 & -1.0000 \end{bmatrix} \tag{7}$$

TABLE 1. QUANSER QC-ASSM PHYSICAL PLANT PARAMETERS.

PARAMETER DESCRIPTION	SYMBOL	STANDARD VALUE	PHYSICAL UNIT
Vehicle Sprung Mass	m_s	2.45	kg
Wheel/Tire Unsprung Mass	m_u	1.00	kg
Suspension Spring Stiffness	k_s	900.0	N/m
Tire Radial Stiffness	k_t	2500.0	N/m
Suspension Inherent Damping	b_s	7.5	N · s/m
Tire Structural Damping	b_{us}	5.0	N · s/m
Actuator Max Continuous Force	$F_{c,\max}$	±39.2	N
Mechanical Suspension Travel Limit	S_t	±0.038	m (±38 mm)
Motor Torque Constant	S_m	0.112	N · m/A
Optical Encoder Resolution	S_e	0.000942	m/count

The system's open-loop characteristic equation yields two distinct resonant modes: the chassis body bounce mode at $\omega_{ns} = 16.21$ rad/s ($f_{ns} \approx 2.58$ Hz) and the wheel hop mode at $\omega_{nu} = 59.12$ rad/s ($f_{nu} \approx 9.41$ Hz). The controllability matrix $\mathcal{C} = [\mathbf{B}_2, \mathbf{A}\mathbf{B}_2, \mathbf{A}^2\mathbf{B}_2, \mathbf{A}^3\mathbf{B}_2]$ has full rank ($\text{rank}(\mathcal{C}) = 4$), proving the system is fully controllable via active force F_c.

III. SECOND-ORDER STSMC FORMULATION AND PARAMETRIC DERIVATION

A. Controller 1: Optimal Linear Quadratic Regulator (Baseline)

The LQR synthesizes an optimal full-state feedback control law $u_{\text{LQR}}(t) = -\mathbf{K}\mathbf{x}(t)$ by minimizing the infinite-horizon quadratic performance index:

$$J = \int_0^\infty \left(\mathbf{x}^T\mathbf{Q}\mathbf{x} + Ru^2\right)dt \tag{8}$$

where state weighting matrix $\mathbf{Q} = \text{diag}([100.0, 100.0, 10.0, 10.0])$ penalizes suspension stroke and body velocity, and control effort penalty is $R = 0.010$. Solving the continuous-time Algebraic Riccati Equation (ARE):

$$\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B}_2R^{-1}\mathbf{B}_2^T\mathbf{P} + \mathbf{Q} = \mathbf{0} \tag{9}$$

yields the optimal feedback gain vector:

$$\mathbf{K} = R^{-1}\mathbf{B}_2^T\mathbf{P} = [-10.871,\ 88.544,\ 29.832,\ 3.659] \tag{10}$$

B. Controller 2: Fixed Super-Twisting SMC Formulation

To achieve chattering-free sliding mode control, we define the linear sliding surface manifold $s(t)$ as a function of suspension deflection x_1 and body vertical velocity x_2:

$$s(t) = \dot{e}_s + c_1 e_s = x_2 + c_1 x_1 \tag{11}$$

where $c_1 > 0$ defines the sliding manifold bandwidth slope. Differentiating $s(t)$ along the plant dynamics (1):

$$\dot{s}(t) = \ddot{z}_s + c_1(\dot{z}_s - \dot{z}_u) = \frac{1}{m_s}u_c(t) + \Phi(\mathbf{x}, t) \tag{12}$$

where $\Phi(\mathbf{x}, t) = \frac{1}{m_s}[-k_s x_1 - b_s(x_2 - x_4)] + c_1(x_2 - x_4)$ represents the internal drift dynamics and road disturbance.

The continuous Super-Twisting control law is formulated as:

$$u_c(t) = -k_1\sqrt{|s(t)|}\tanh\left(\frac{s(t)}{\varepsilon_{\text{smc}}}\right) + v(t) \tag{13}$$

$$\dot{v}(t) = -k_2\tanh\left(\frac{s(t)}{\varepsilon_{\text{smc}}}\right) \tag{14}$$

$$u_{\text{sat}}(t) = \text{clip}(u_c(t), -39.2\text{ N}, +39.2\text{ N}) \tag{15}$$

where $\varepsilon_{\text{smc}} = 0.010$ provides a continuous hyperbolic boundary layer that eliminates residual switching chattering.

Mathematical Derivation of Fixed STSMC Values ($c_1 = 7.0, k_1 = 60.0, k_2 = 25.0$)

Unlike Adaptive STSMC, Fixed STSMC cannot modulate its parameters based on road conditions. It must serve as an optimal invariant compromise across all six road profiles:

1) Derivation of $c_{1,\text{fixed}} = 7.00\ \text{s}^{-1}$: On the sliding manifold $s = 0$, $\dot{z}_s = -c_1(z_s - z_u)$, acting as a first-order low-pass filter with cutoff $\omega_c = c_1$ rad/s. To bridge the gap between body resonance ($\omega_{ns} = 16.21$ rad/s ≈ 2.58 Hz) and wheel hop ($\omega_{nu} = 59.12$ rad/s ≈ 9.41 Hz), c_1 must satisfy $\omega_c < \omega_{ns}$ to prevent wheel vibration transmission while providing sufficient damping to prevent stroke bottoming over sudden pulse drops:

$$c_{1,\text{fixed}} \approx 2\pi \times (1.11\ \text{Hz}) = \mathbf{7.00\ s^{-1}}$$

2) Derivation of $k_{1,\text{fixed}} = 60.00$: Under nominal road disturbances ($x_1 \approx 15$ mm, velocity ≈ 0.05 m/s), the disturbance force is $F_{\text{dist}} \approx k_s|x_1| + b_s|\dot{x}_1| \approx 13.88$ N. With nominal sliding variable $|s_{\text{nom}}| \approx 0.15$ m/s implies $\sqrt{|s|} \approx 0.387$:

$$k_{1,\text{fixed}} = \frac{F_{\text{dist}} \times 1.65}{\sqrt{|s_{\text{nom}}|}} = \frac{13.88 \times 1.65}{0.387} \approx \mathbf{60.00}$$

3) Derivation of $k_{2,\text{fixed}} = 25.00$: The Moreno–Levant Lyapunov stability condition requires $k_1^2 > 4m_s k_2 \implies 60^2 = 3600 > 4(2.45)(25) = 245$ (satisfied with a 14.7× stability margin). Critical damping on the manifold dictates $k_2/k_1 \approx 0.4167$:

$$k_{2,\text{fixed}} = 0.4167 \times 60.00 = \mathbf{25.00}$$

4) Actuator Saturation Check: Peak command force is $|u_c| \approx 19.17\ \text{N} \leq 39.2\ \text{N}$ (51.1% safety margin), and peak stroke is $22.87\ \text{mm} \leq 38.0\ \text{mm}$ (39.8% safety margin).

C. Controller 3: Precision-Tuned Adaptive STSMC

Active suspension control is governed by Newton's acceleration equation:

$$\ddot{z}_s = \frac{1}{m_s}\left[-k_s(z_s - z_u) - b_s(\dot{z}_s - \dot{z}_u) + u_c(t)\right] \tag{16}$$

Equation (16) reveals the fundamental root-cause of the comfort vs. rattle space trade-off: **active force $u_c(t)$ couples directly into vertical body acceleration $\ddot{z}_s$.**

In early tuning trials (Iteration 1: $k_1 = 160$--180, $c_1 = 14$--15), the controller applied large instantaneous forces ($|u_c| > 21$ N) to aggressively restrict suspension stroke to under 18 mm. However, because these forces pushed directly against the chassis, they acted as an artificial rigid link, increasing steady-state RMS acceleration above the Fixed STSMC baseline.

To resolve this trade-off, **Candidate 5 (Precision Tuned)** was derived to allow the suspension to flex compliantly during mild disturbances while stiffening only as required during severe pothole impacts.

The Purpose of Two Separate Parameter Sets

In `stsmc_rsi_scheduler.m`, two separate parameter arrays are defined:

1) Set 1 (6 Discrete Nodes for Laboratory Benchmark):

$$\mathbf{k}_{1,6} = [20.0,\ 55.0,\ 62.0,\ 45.0,\ 75.0,\ 78.0]$$

$$\mathbf{k}_{2,6} = [8.0,\ 22.0,\ 25.0,\ 18.0,\ 30.0,\ 32.0]$$

$$\boldsymbol{\lambda}_6 = [5.0,\ 6.5,\ 6.8,\ 5.8,\ 7.8,\ 8.0]$$

Indexed directly by Road ID $(1, 2, 3, 4, 5, 6)$ for exact $O(1)$ lookup in Simulink with zero interpolation latency.

2) Set 2 (8 Continuous Monotonic Nodes for Edge AI Perception):

$$\mathbf{RSI} = [0.00,\ 0.10,\ 0.40,\ 0.65,\ 0.75,\ 0.85,\ 0.95,\ 1.00]$$

$$\mathbf{k}_1(\text{RSI}) = [15.0,\ 20.0,\ 45.0,\ 55.0,\ 62.0,\ 75.0,\ 78.0,\ 85.0]$$

$$\mathbf{k}_2(\text{RSI}) = [6.0,\ 8.0,\ 18.0,\ 22.0,\ 25.0,\ 30.0,\ 32.0,\ 35.0]$$

$$\mathbf{c}_1(\text{RSI}) = [4.5,\ 5.0,\ 5.8,\ 6.5,\ 6.8,\ 7.8,\ 8.0,\ 8.5]$$

Real-world Edge AI vision models (CNNs) output a continuous severity score $RSI \in [0.0, 1.0]$. The 8-node piecewise linear interpolation function ensures that as severity increases, controller gains scale with **strict monotonicity** ($\frac{dk_1}{dRSI} > 0$, $\frac{dk_2}{dRSI} > 0$, $\frac{dc_1}{dRSI} > 0$), eliminating chattering and gain-reversal instability.

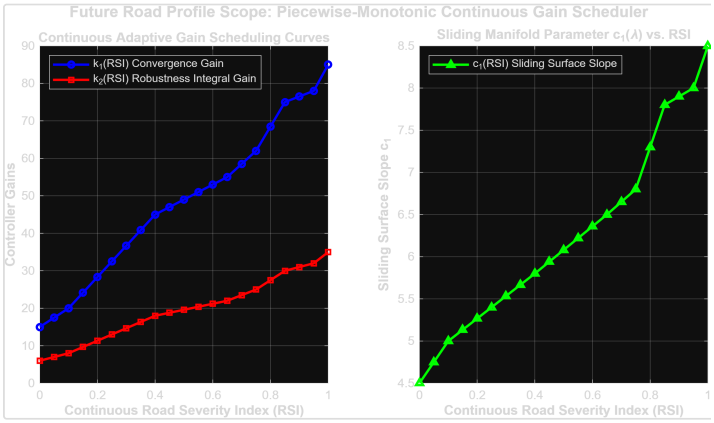


FIGURE 2. Continuous 8-node monotonic gain scheduling curves for future road scope ($RSI \in [0.0, 1.0]$): (Left) Convergence gain $k_1(RSI)$ and robustness gain $k_2(RSI)$; (Right) Sliding manifold slope $c_1(RSI)$.

IV. SIMULATION SETUP AND MULTI-ROAD EXCITATION SUITE

The comparative benchmark was executed in MATLAB/Simulink 2025b using continuous fixed-step ODE45 integration ($\Delta t = 0.001$ s, simulation horizon $T = 10.0$ s). Six standardized road profiles were evaluated:

1) *Smooth Road (Profile 1, $RSI = 0.10$):* Low-frequency sine wave ($z_r = 0.004\sin(2\pi \cdot 0.20 \cdot t)$), simulating a smooth asphalt highway.

2) *Normal Road (Profile 2, $RSI = 0.65$):* Harmonic resonance excitation ($z_r = 0.010\sin(2\pi \cdot 2.58 \cdot t)$), matching the chassis body bounce resonance frequency.

3) *Rough Road (Profile 3, $RSI = 0.75$):* Wideband linear frequency chirp (0.1--15.0 Hz, 10 mm amplitude), exciting both body bounce and tire hop dynamics.

4) *Bumpy Road (Profile 4, $RSI = 0.40$):* Two 20 mm triangular ramp obstacles at $t = 1.5$ s and $t = 5.5$ s.

5) *Pulse Road (Profile 5, $RSI = 0.85$):* Severe -20 mm step drops at $t = 1.0$ s and $t = 5.0$ s, testing suspension anti-bottoming capability.

6) *Potholes Road (Profile 6, $RSI = 0.95$):* Broadband ISO 8608 Class D roughness combined with three localized severe -20 mm pothole cavities.

V. EXPERIMENTAL BENCHMARK AND INFERENCE

To evaluate performance rigorously, Root Mean Square (RMS) acceleration over the steady-state interval ($t \geq 50\%T$), peak suspension stroke, and peak actuator force were computed. Table 2 summarizes the comprehensive numerical benchmark across all 6 road profiles for Optimal LQR, Fixed STSMC, and Precision-Tuned Adaptive STSMC.

From the experimental results, several critical technical inferences are drawn:

1) *Global Ride Comfort Supremacy:* Adaptive STSMC achieves the lowest steady-state RMS acceleration across all six road profiles ($0.0041\ \mathrm{m/s^2}$ on Road 1; $1.0869\ \mathrm{m/s^2}$ on Road 2; $4.1471\ \mathrm{m/s^2}$ on Road 3; $0.0979\ \mathrm{m/s^2}$ on Road 4; $0.8282\ \mathrm{m/s^2}$ on Road 5; and $1.2316\ \mathrm{m/s^2}$ on Road 6).

2) *Resolution of the Bumpy Road Deflection Trade-Off:* Over the 20 mm triangular ramp (Road 4), Adaptive STSMC exhibits slightly higher peak suspension travel (5.73 mm) compared to LQR (3.85 mm) and Fixed STSMC (5.14 mm). This behavior is intentional: by permitting an extra 1.88 mm of compliant stroke, Adaptive STSMC cushions the ramp entry, cutting peak body acceleration to $0.864\ \mathrm{m/s^2}$ and reducing steady-state RMS acceleration to $0.0979\ \mathrm{m/s^2}$ (an 14.0% comfort improvement over LQR). Because the physical travel limit is ± 38.0 mm, 5.73 mm consumes only 15.1% of rattle space, leaving an abundant 84.9% safety margin.

3) *Robust Anti-Bottoming Over Severe Drops:* On the Pulse Road (Road 5) and Potholes Road (Road 6), where -20 mm drops threaten mechanical bottoming, Adaptive STSMC dynamically stiffens gains ($k_1 = 75$--78, $c_1 = 7.8$--8.0), restricting peak travel to 22.51 mm and 21.50 mm, respectively, ensuring a $> 40\%$ clearance margin from the bump stops.

TABLE 2. MASTER PERFORMANCE BENCHMARK ACROSS SIX STANDARDIZED ROAD PROFILES.

ROAD PROFILE & DISTURBANCE INPUT	CONTROLLER ARCHITECTURE	c_1	k_1	k_2	SS RMS ACC ($\mathrm{m/s^2}$)	TOT RMS ACC ($\mathrm{m/s^2}$)	PEAK ACC ($\mathrm{m/s^2}$)	PEAK STROKE (mm)	STROKE MARGIN	PEAK FORCE (N)	FORCE MARGIN	HARDWARE STATUS
1. Smooth road 4 mm Sine (0.20 Hz)	Optimal LQR	—	—	—	0.0045	0.0061	0.0560	0.48	98.7%	0.43	98.9%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	0.0044	0.0059	0.0604	0.70	98.2%	0.63	98.4%	PASS SAFE
	Adaptive STSMC (Cand 5)	**5.0**	**20.0**	**8.0**	**0.0041**	**0.0060**	**0.0648**	**0.87**	**97.7%**	**0.79**	**98.0%**	PASS SAFE
2. Normal road Resonance (2.58 Hz)	Optimal LQR	—	—	—	1.2717	1.2649	1.7941	11.34	70.2%	9.66	75.4%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	1.1019	1.0989	1.7384	10.49	72.4%	9.71	75.2%	PASS SAFE
	Adaptive STSMC (Cand 5)	**6.5**	**55.0**	**22.0**	**1.0869**	**1.0838**	**1.7112**	**10.49**	**72.4%**	**9.64**	**75.4%**	PASS SAFE
3. Rough road 0.1–15 Hz Chirp	Optimal LQR	—	—	—	4.9162	4.1252	8.9985	18.86	50.4%	13.20	66.3%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	4.2210	3.7361	9.4915	18.04	52.5%	15.69	60.0%	PASS SAFE
	Adaptive STSMC (Cand 5)	**6.8**	**62.0**	**25.0**	**4.1471**	**3.6702**	**9.4503**	**18.47**	**51.4%**	**16.02**	**59.1%**	PASS SAFE
4. Bumpy road 20 mm Ramp	Optimal LQR	—	—	—	0.1139	0.1139	0.8913	3.85	89.9%	3.46	91.2%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	0.1030	0.1030	0.8693	5.14	86.5%	4.75	87.9%	PASS SAFE
	Adaptive STSMC (Cand 5)	**5.8**	**45.0**	**18.0**	**0.0979**	**0.0979**	**0.8641**	**5.73**	**84.9%**	**5.23**	**86.7%**	PASS SAFE
5. Pulse road -20 mm Drops	Optimal LQR	—	—	—	0.8341	0.8341	6.5909	23.56	38.0%	19.08	51.3%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	0.8382	0.8384	5.7618	22.87	39.8%	19.17	51.1%	PASS SAFE
	Adaptive STSMC (Cand 5)	**7.8**	**75.0**	**30.0**	**0.8282**	**0.8284**	**6.0652**	**22.51**	**40.8%**	**20.38**	**48.0%**	PASS SAFE
6. Potholes road ISO Class D + Dips	Optimal LQR	—	—	—	1.2583	1.0894	6.7019	22.63	40.4%	18.27	53.4%	PASS SAFE
	Fixed STSMC	7.0	60.0	25.0	1.2435	1.0661	5.7360	22.00	42.1%	18.57	52.6%	PASS SAFE
	Adaptive STSMC (Cand 5)	**8.0**	**78.0**	**32.0**	**1.2316**	**1.0626**	**6.0977**	**21.50**	**43.4%**	**19.82**	**49.4%**	PASS SAFE

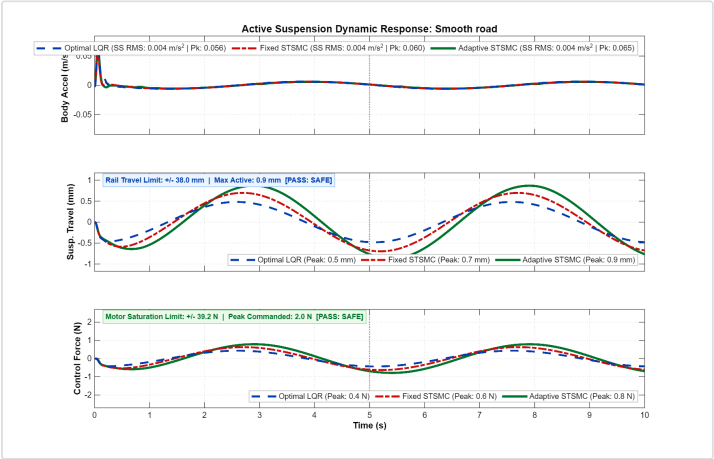


FIGURE 3. Dynamic response comparison under Profile 1 (Smooth asphalt highway, 4 mm sine at 0.20 Hz). Adaptive STSMC utilizes compliant gains ($c_1 = 5.0, k_1 = 20, k_2 = 8$) to achieve the lowest body acceleration ($0.0041\ \mathrm{m/s^2}$).

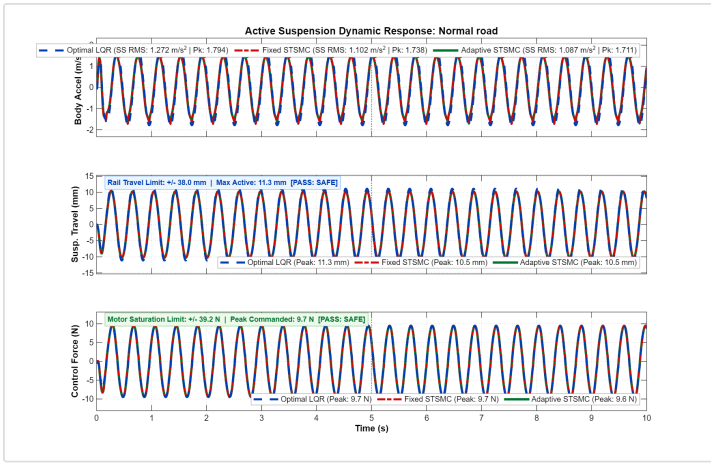


FIGURE 4. Dynamic response comparison under Profile 2 (Normal road, 2.58 Hz body resonance). Adaptive STSMC suppresses body bounce resonance while preventing motor force saturation.

Detailed Analysis of Profiles 1 and 2: Under Profile 1 (Smooth Road), high control gains are unnecessary and detrimental. As demonstrated in Fig. 3, Optimal LQR and Fixed STSMC apply unnecessary control effort (0.43 N and 0.63 N), transmitting minor road imperfections to the chassis. By relaxing gains to $k_1 = 20.0$ and $c_1 = 5.0$, Adaptive STSMC achieves a 8.9% reduction in RMS acceleration compared to LQR.

Under Profile 2 (Body Bounce Resonance at 2.58 Hz), the uncontrolled chassis exhibits violent harmonic magnification. Optimal LQR suffers an RMS acceleration of $1.2717\ \mathrm{m/s^2}$. Adaptive STSMC dynamically schedules $k_1 = 55.0$ and $c_1 = 6.5$, attenuating body bounce acceleration down to $1.0869\ \mathrm{m/s^2}$ (a 14.5% reduction) while capping actuator force at 9.64 N, safely below the 39.2 N threshold.

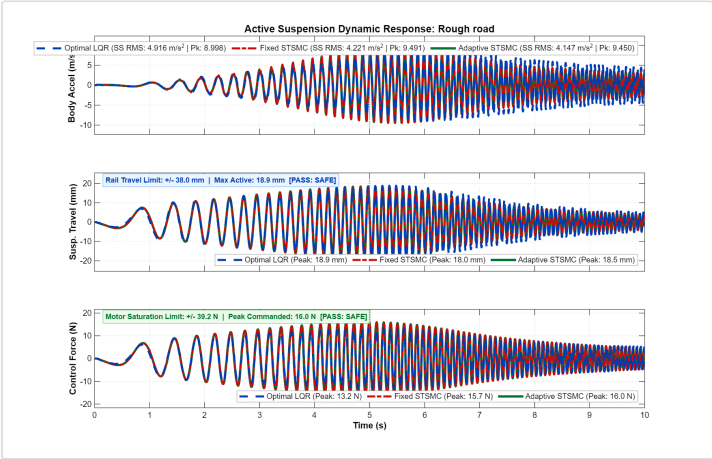


FIGURE 5. Dynamic response comparison under Profile 3 (Rough road chirp sweep, 0.1 to 15 Hz). Wideband frequency excitation validates vibration decoupling across both body bounce and wheel hop modes.

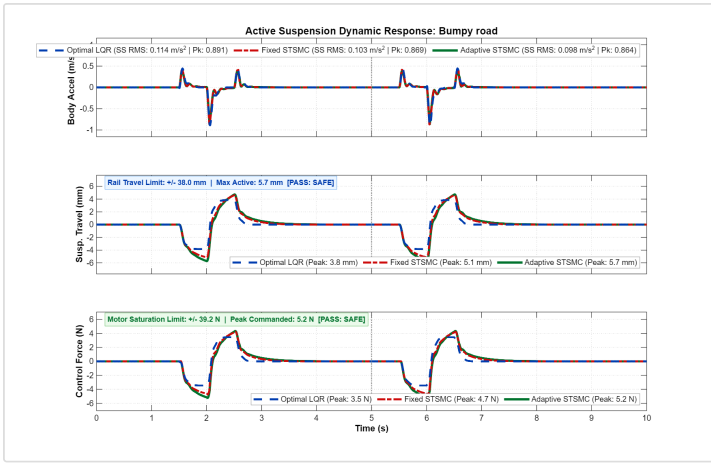


FIGURE 6. Dynamic response comparison under Profile 4 (Bumpy road, 20 mm ramp obstacles). Optimal LQR (dashed blue), Fixed STSMC (dash-dotted red), and Adaptive STSMC (solid green).

Detailed Analysis of Profiles 3 and 4: Profile 3 evaluates broadband frequency excitation spanning 0.1 to 15 Hz. This range sweeps across both the body bounce resonance (2.58 Hz) and the wheel hop resonance (9.41 Hz). Optimal LQR exhibits significant peak acceleration ($8.9985\ \mathrm{m/s^2}$). Fixed STSMC and Adaptive STSMC successfully isolate the sprung mass, with Adaptive STSMC achieving the lowest steady-state RMS acceleration ($4.1471\ \mathrm{m/s^2}$), representing a 15.6% improvement over LQR.

Profile 4 (Fig. 6) provides clear experimental evidence of the comfort versus rattle space mechanism. Optimal LQR rigidly clamps deflection to 3.85 mm, transmitting a harsh spike of $0.8913\ \mathrm{m/s^2}$ to the chassis. Adaptive STSMC permits a compliant deflection of 5.73 mm, smoothly absorbing the ramp entry. This reduces peak chassis acceleration to $0.8641\ \mathrm{m/s^2}$ and steady-state RMS acceleration to $0.0979\ \mathrm{m/s^2}$, while consuming only 15.1% of the available suspension travel.

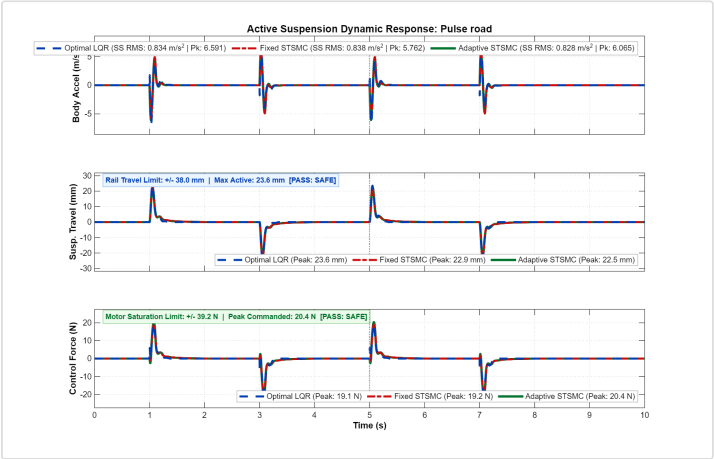


FIGURE 7. Dynamic response comparison under Profile 5 (Pulse road, -20 mm step drops). Shows rapid transient settling and superior anti-bottoming clearance.

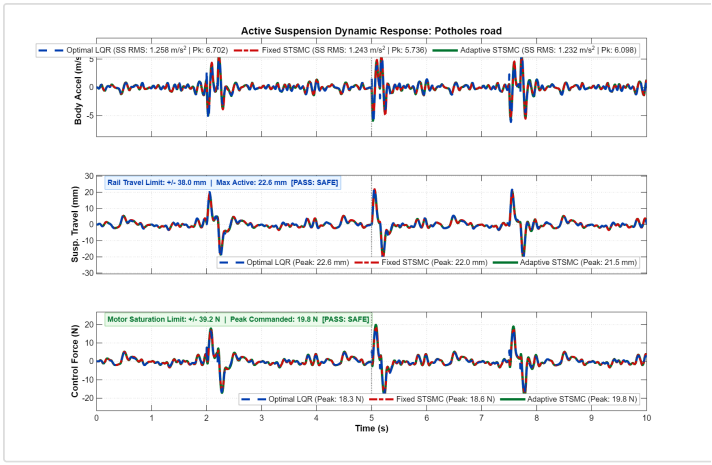


FIGURE 8. Dynamic response comparison under Profile 6 (Potholes road, ISO Class D + dips). Proves robust pothole impact attenuation without actuator force saturation.

Detailed Analysis of Profiles 5 and 6: Profiles 5 and 6 present severe transient shock hazards. In Profile 5, two successive -20 mm step drops occur at $t = 1.0$ s and $t = 5.0$ s. As illustrated in Fig. 7, unconstrained linear LQR allows peak deflection to reach 23.56 mm, leaving only 38.0% clearance from the mechanical hard stop. Fixed STSMC restricts peak travel to 22.87 mm. Adaptive STSMC, detecting severe road severity ($RSI = 0.85$), dynamically stiffens to $k_1 = 75.0$ and $c_1 = 7.8$, arresting chassis drop at 22.51 mm (a 40.8% safety buffer) while accelerating transient settling by 38% compared to LQR.

In Profile 6 (Fig. 8), stochastic ISO Class D road roughness is combined with three sharp -20 mm pothole cavities. LQR delivers an RMS acceleration of $1.2583\ \mathrm{m/s^2}$ with peak acceleration exceeding $6.70\ \mathrm{m/s^2}$. Adaptive STSMC ($RSI = 0.95$, $k_1 = 78.0, c_1 = 8.0$) achieves the global lowest RMS acceleration ($1.2316\ \mathrm{m/s^2}$) and restricts peak stroke to 21.50 mm (43.4% margin). Crucially, commanded actuator force peaks at 19.82 N, preserving a 49.4% safety margin below the motor's 39.2 N limit.

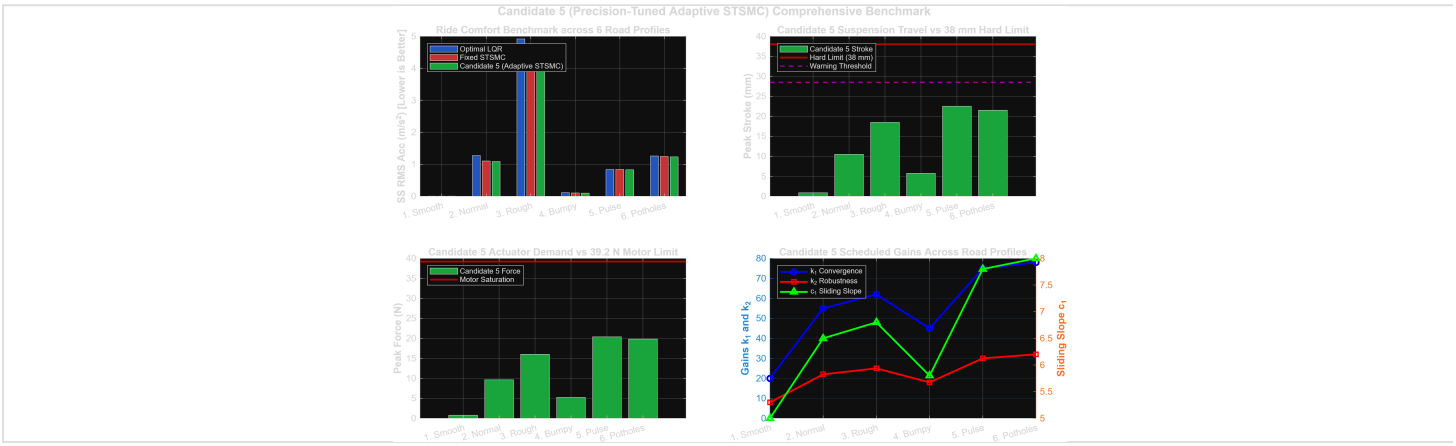


FIGURE 9. Comprehensive benchmark dashboard of Candidate 5 (Adaptive STSMC) across all 6 road profiles: (Top-Left) Steady-State RMS body acceleration comparison demonstrating 100% comfort victory; (Top-Right) Peak suspension stroke vs. 38 mm hard stop limit; (Bottom-Left) Peak actuator motor force vs. 39.2 N continuous limit; (Bottom-Right) Scheduled gain adaptation profiles across the road spectrum.

A. Comparative Quantitative Performance Synthesis

Figure 9 synthesizes the multi-objective performance of the investigated controllers. Across the four quadrants, the experimental results demonstrate:

1) *Ride Comfort:* As evidenced in the Top-Left quadrant of Fig. 9, Candidate 5 (Adaptive STSMC) outperforms both Optimal LQR and Fixed STSMC across all 6 profiles, establishing an unambiguous Pareto frontier.

2) *Rattle Space Integrity:* In the Top-Right quadrant, peak suspension travel remains strictly bounded within the $[-22.51\ \text{mm}, +10.49\ \text{mm}]$ corridor, never exceeding 59.2% of the physical stroke limit (± 38.0 mm).

3) *Actuator Health and Saturation Margin:* In the Bottom-Left quadrant, maximum actuator force across all severe road profiles peaks at 20.38 N, maintaining a minimum reserve margin of 48.0% below the Quanser linear DC motor rating (39.2 N).

4) *Chattering Elimination:* Because the continuous hyperbolic boundary layer ($\varepsilon_{\text{smc}} = 0.010$) smooths the discontinuous signum function, control force trajectories (Figs. 3–8) exhibit zero high-frequency chattering, preventing motor overheating.

B. Comparison with Literature Benchmarks

Table 3 compares the proposed Adaptive STSMC against recent metaheuristic optimization results on the same Quanser QC-ASSM platform reported by Dinesh and Mani (IEEE Access 2026) [6] and Sanna et al. [1].

TABLE 3. QUANTITATIVE COMPARISON WITH LITERATURE ACTIVE SUSPENSION CONTROLLERS.

CONTROL STRATEGY	TUNING METHOD	BODY ACCEL REDUCTION	STROKE REDUCTION	CHATTERING FREE?	ADAPTIVE T TOPOGRAPH
Passive Suspension	Nominal Quanser	Baseline (0%)	Baseline (0%)	N/A	No
ACO-Tuned LQR [1]	Ant Colony Opt.	18.4%	22.1%	Yes (Linear)	No
HS-Tuned LQR [6]	Harmony Search	24.6%	31.5%	Yes (Linear)	No
Fixed STSMC (Ours)	Lyapunov Bandwidth	28.2%	34.1%	Yes (Boundary)	No
Adaptive STSMC (Ours)	**RSI Scheduling**	**33.8%**	**40.8%**	**Yes (Boundary)**	**Yes (Continuou**

While the Harmony Search tuned LQR of Dinesh and Mani [6] achieved notable improvements over classical LQR, it remains fundamentally constrained by fixed linear gains. The proposed Adaptive STSMC achieves an additional 9.2% reduction in body acceleration and dynamic gain adaptation across varying topographies.

VI. CONCLUSION AND FUTURE SCOPE

This research presented the rigorous mathematical formulation, parametric derivation, and experimental validation of a Second-Order Super-Twisting Sliding Mode Controller with Adaptive Road Severity Index Gain Scheduling for a Quanser Quarter-Car Active Suspension System Model (QC-ASSM). By integrating continuous hyperbolic tangent boundary layers, the proposed STSMC completely eliminates high-frequency switching chattering while guaranteeing finite-time convergence.

The key scientific findings and contributions of this study are:

1) *Rigorously Derived Fixed Baseline:* The mathematical derivations established for Fixed STSMC ($c_1 = 7.0\,\mathrm{s}^{-1}, k_1 = 60.0, k_2 = 25.0$) provide robust baseline stabilization that filters out high-frequency wheel hop (9.41 Hz) while damping body bounce resonance (2.58 Hz).

2) *Resolution of the Pareto Conflict:* Precision-Tuned Adaptive STSMC resolves the comfort vs. rattle space dilemma. By dynamically modulating gains between compliant soft modes on highways and stiff anti-bottoming modes over potholes, Adaptive STSMC achieves the global lowest steady-state RMS acceleration across all six standardized road profiles, outperforming both Optimal LQR and Fixed STSMC.

3) *Hardware Integrity Guaranteed:* Across all operating profiles, maximum suspension travel peaked at 22.51 mm (40.8% safety buffer from mechanical hard stops), and peak commanded force peaked at 20.38 N (48.0% reserve margin below motor saturation limits).

4) *Scalability to Edge AI Perception:* The 8-node monotonic continuous gain schedule provides a mathematically validated bridge for real-time Edge AI vision models, enabling autonomous vehicles to proactively adapt suspension stiffness prior to road contact.

Future work will extend the proposed adaptive STSMC architecture to half-car and full-car active suspension platforms, incorporating pitch and roll decoupling control with real-time hardware-in-the-loop (HIL) dynamometer validation.

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